

Today

Logarithmic + Exponential Growth

Reminders

- HW 3 Due Tomorrow
- Exam 1 Monday
 - Calculator, pencil
 - Be familiar w/ Eqn sheet (I will print)

Bisection requires $\log_2\left(\frac{\Delta x}{\epsilon_s}\right)$ function calls
 usually c flops

Algorithm 1

Pick up end of strip
 for $i = 1$ to L/ϵ_s
 Fold one section (accordion style)
 end

time	times executed
t_p	1
t_f	$\lceil L/\epsilon_s \rceil$

runtime
 $t_p + t_f \lceil L/\epsilon_s \rceil$
 $t_p + t_f \frac{L}{\epsilon_s} = O(L)$

Algorithm 2

1 Pick up end of strip
 2 $l_t \leftarrow L$
 3 while $l_t > \epsilon_s$
 4 ~~Run to other end~~
 5 Pick up end and fold in half
 6 $l_t \leftarrow l_t/2$
 7 end

time	times executed
t_p	1
0	1
t_f	$\log_2\left(\frac{L}{\epsilon_s}\right)$
0	

Iter	l_t
0	L
1	$\frac{L}{2}$
2	$\frac{L}{4}$
k	$\frac{L}{2^k}$

$\epsilon_s = l_t = \frac{L}{2^k}$
 $2^k = \frac{L}{\epsilon_s}$
 $k = \log_2\left(\frac{L}{\epsilon_s}\right)$

runtime:
 $t_p + t_f \log_2\left(\frac{L}{\epsilon_s}\right)$
 $= O(\log(L))$

Algorithm 2 (Running)

1 Pick up end of strip
 2 $l_t \leftarrow L$
 3 while $l_t > \epsilon_s$
 4 Run to other end
 5 Pick up end
 6 $l_t \leftarrow l_t/2$
 7 end

time	times executed
t_p	1
0	
l_t/v	$\log\left(\frac{L}{\epsilon_s}\right)$
t_p	$\log\left(\frac{L}{\epsilon_s}\right)$

runtime:
 $t_p + \sum_{j=1}^{\log_2(L/\epsilon_s)} \left(\frac{L}{2^j} \frac{1}{v} + t_p \right)$
 $= t_p \left(1 + \log_2\left(\frac{L}{\epsilon_s}\right) \right) + \sum_{j=1}^{\log_2(L/\epsilon_s)} \frac{L}{2^j} \frac{1}{v}$
 $\leq t_p \left(1 + \log_2\left(\frac{L}{\epsilon_s}\right) \right) + \frac{L}{v} \sum_{j=1}^{\infty} 2^{-j}$
 $= t_p \left(1 + \log_2\left(\frac{L}{\epsilon_s}\right) \right) + \frac{2L}{v}$
 $= O(L)$

$v = 10m/s$

Alg 1 and Alg 2(R) both have $O(L)$ runtime
 $\downarrow$ $\downarrow$
 $\frac{t_f}{\epsilon_s} L$ $\frac{2}{v} L$

$\frac{\epsilon_s}{t_f} = \frac{1/3 m}{1 s}$
 $\frac{t_f}{\epsilon_s} = 3$

$\frac{2}{v} = 0.2$

Alg 2 (R)
 takes $\sim 15 \times$ Alg

Logarithmic Growth

$O(1)$ $O(\log(n))$ $O(n^{\epsilon})$ $O(n^{1/2})$ $O(n)$ $O(n^2)$ $O(n^3)$
 $\log(n)$ grows slower asymptotically than any polynomial n^ϵ $\epsilon > 0$ $\nearrow O(n^\epsilon)$

$$\begin{matrix} f(n) = O(n) \\ \uparrow \\ f(n) \in O(n) \end{matrix} \rightarrow O(\log(n)) \subsetneq O(n^\epsilon) \quad \forall \epsilon > 0$$

↑
strict subset

For sets of functions $O(f(n))$ and $O(g(n))$: $O(f(n)) \subsetneq O(g(n))$ means
 1. $O(g(n))$ contains every function in $O(f(n))$
 2. $O(g(n))$ contains at least 1 function **NOT** in $O(f(n))$

How can we prove this

$$\lim_{n \rightarrow \infty} \frac{f(n)}{g(n)} = 0 \Rightarrow O(f(n)) \subsetneq O(g(n))$$

$$\lim_{n \rightarrow \infty} \frac{\log(n)}{n^\epsilon} \stackrel{\text{L'Hopital}}{=} \lim_{n \rightarrow \infty} \frac{n^{-1}}{\epsilon n^{\epsilon-1}} = \lim_{n \rightarrow \infty} \frac{1}{\epsilon n^\epsilon} = 0$$

$$\therefore O(\log(n)) \subsetneq O(n^\epsilon) \quad \forall \epsilon > 0$$

$$\log_a(n) = \frac{\log_b(n)}{\log_b(a)} \quad \leftarrow \text{constant}$$

$$\therefore O(\log_a(n)) = O(\log_b(n)) \quad \forall a, b > 0$$

Exponential Growth

$a^n, a > 1$ grows faster than any polynomial

$$e^n = \sum_{j=0}^{\infty} \frac{n^j}{j!} \quad a^n = \sum_{j=0}^{\infty} \frac{(\log(a))^j n^j}{j!} \geq \frac{(\log(a))^{k+1} n^{k+1}}{(k+1)!}$$

$$\lim_{n \rightarrow \infty} \frac{n^k}{a^n} \leq \lim_{n \rightarrow \infty} \frac{(k+1)!}{(\log(a))^{k+1}} \cdot \frac{n^k}{n^{k+1}} = \lim_{n \rightarrow \infty} \frac{1}{n} = 0$$

$$\therefore O(n^k) \subsetneq O(a^n) \quad \forall a > 1, k > 0$$

Let $1 < a < b$

$$O(a^n) \subsetneq O(b^n) \quad \forall 1 < a < b$$

Factorial Growth

$n!$
 $= n(n-1)(n-2)\dots 1$ grows faster than any exponential

$$\lim_{n \rightarrow \infty} \frac{a^n}{n!} = \lim_{n \rightarrow \infty} \frac{\overbrace{a \cdot a \cdot a \dots a}^N \cdot \overbrace{a \cdot a \dots a}^{n-N}}{\underbrace{1 \cdot 2 \dots N}_N \cdot \underbrace{(N+1)(N+2) \dots n}_{n-N}} \leq \lim_{n \rightarrow \infty} \frac{a^N}{N!} \left(\frac{1}{2}\right)^{n-N} = 0$$

$N = 2a$

$$O(a^n) \subsetneq O(n!) \quad \forall a > 1$$

$$O(1) \subsetneq O(\log(n)) \subsetneq O(n) \subsetneq O(n^k) \subsetneq O(a^n) \subsetneq O(n!)$$

n	$\log_2 n$	n	n^3	2^n	time	$n!$	time
10	3.3	10	10^3	10^3	μs	3.6×10^6	ms
20	4.3	20	8×10^3	10^6	ms	2.4×10^{18}	77 yr
30	4.9	30	2.7×10^4	10^9	1 s	2.7×10^{32}	10^{15} yr
60	5.9	60	2×10^5	10^{18}	36 yr	8×10^{81}	10^{65} yr

Times assume one operation per flop at 10^9 flop/s.

$\det(A)$ with Laplace Expansion requires $n!$ flops

Cramer's $A^{-1} = \frac{\text{adj}(A)}{\det(A)}$ is extremely inefficient